

Binomial Probability

NOTE: Unless otherwise specified, leave your answers in unsimplified form.

1. Assume that the probability that a child is female is $\frac{1}{2}$, and that sex is independent from child to child. Giving your answers as fractions in simplest form, find the probability that in a family of five children:
 - a. All are boys
 - b. There are two girls and three boys
 - c. There are four boys and one girl
 - d. At least one will be a girl
2. In a one-day cricket game, a batsman has a chance of $\frac{1}{5}$ of hitting a boundary every time he faces a ball. If he faces all six balls in an over, what is the probability that he will hit exactly two boundaries, assuming that successive strikes are independent?
3. During the wet season, it rains on average every two days out of three. Let W denote the number of wet days in a week. Assuming that the types of weather on successive days are independent events, find:
 - a. $P(W = 3)$
 - b. $P(\bar{W} = 2)$
 - c. $P(W = 0)$
 - d. $P(W \geq 1)$
4. A marksman finds that on average he hits the target five times out of six. Assuming that successive shots are independent events, find the probability that in four shots:
 - a. He has exactly three hits
 - b. He has exactly two misses
5. A jury roll contains 200 names, 70 of females and 130 of males. If twelve jurors are randomly selected, what is the probability of ending up with an all-male jury?
6. A die is rolled twelve times. Find the probability that 5 appears on the uppermost face:
 - a. Exactly three times
 - b. Exactly eight times
 - c. Ten or more times (that is ten, eleven or twelve times)
7. A die is rolled six times. Let N denote the number of times that the number 3 is shown on the uppermost face. Find, correct to four decimal places:
 - a. $P(N = 2)$
 - b. $P(N < 2)$
 - c. $P(N \geq 2)$
8. Five out of six people surveyed think that Tasmania is the most beautiful state in Australia. What is the probability that in a group of 15 randomly selected people, at least 13 of them think that Tasmania is the most beautiful state in Australia?
9. An archer finds that on average he hits the bulls-eye nine times out of ten. Assuming that successive attempts are independent, find the probability that in twenty attempts:
 - a. He scores at least eighteen hits
 - b. He misses at least once
10. A torch manufacturer finds that on average 9% of the bulbs are defective. What is the probability that in a randomly selected batch of ten one-bulb torches:
 - a. There will be no more than two with defective bulbs
 - b. There will be at least two with defective bulbs
11. A poll indicates that 55% of people support Labor party policy. If five people are selected at random, what is the probability that a majority of them will support Labor party policy? Give your answer correct to three decimal places.
12. In a multiple-choice test, there are ten questions, and each questions has five possible answers, only one of which is correct. What is the probability of answering exactly seven questions correctly by chance alone? Give your answer correct to six significant figures.

13. The probability that a small earthquake occurs somewhere in the world on any one day is 0.95.
Assuming that earthquake frequencies on successive days are independent, what is the probability that a small earthquake occurs on exactly 28 of January's 31 days? Leave your answer in index form.
14. The probability that a jackpot prize will be won in a given lottery is 0.012.
 - a. Find, correct to five decimal places, the probability that the jackpot prize will be won:
 - i. Exactly once in ten independent lottery draws
 - ii. At least once in ten independent lottery draws.
 - b. The jackpot prize is initially \$10000 and increases by \$10000 each time the prize is not won. Find, correct to five decimal places, the probability that the jackpot prize will exceed \$200000 when it is finally won.
15. a. How many times must a die be rolled so that the probability of rolling at least one six is greater than 95%?
b. How many times must a coin be tossed so that the probability of tossing at least one tail is greater than 99%?
16. Five families have three children each.
 - a. Find, correct to three decimal places, the probability that:
 - i. At least one of these families has three boys
 - ii. Each family has more boys than girls.
 - b. What assumptions have been made in arriving at your answer?
17. Comment on the validity of the following arguments:
 - a. 'In the McLaughlin Library, 10% of the books are mathematics books. Hence if I go to a shelf and choose five books from that shelf, then the probability that all five books are mathematics books is 10^{-5} .'
 - b. During an election, 45% of voters voted for party A. Hence if I select a street at random, and then select a voted from each of four houses in the street, the probability that exactly two of those voters voted for party A is ${}^4C_2 \times (0.45)^2 \times (0.55)^2$.
18. During winter it rains on average 18 out of 30 days. Five winter days are selected at random. Find, correct to four decimal places, the probability that:
 - a. The first two days chosen will be fine and the remainder wet,
 - b. More rainy days than fine days have been chosen.
19. A tennis player finds that on average he gets his serve in eight out of every ten attempts and serves an ace once every fifteen serves He serves four times. Assuming that successive serves are independent events, find, correct to six decimal places, the probability that:
 - a. All four serves are in
 - b. He hits at least three aces
 - c. He hits exactly three aces and the other serve lands in
20. A man is restoring ten old cars, six of the, manufactured in 1955 and four of them manufactured in 1962. When he tries to start them, on average the 1955 models will start 65% of the time and the 1962 models will start 80% of the time. Find, correct to four decimal places, the probability that at any time:
 - a. Exactly three of the 1955 models and one of the 1962 models will start,
 - b. Exactly four of the cars will start. [HINT: You will need to consider five cases]
21. An apple exporter deals in two types of apples, Red Delicious and Golden Delicious. The ratio of Red Delicious to Golden Delicious is 4 : 1. The apples are randomly mixed together before they are boxed. One in every fifty Golden Delicious and one in every one hundred Red Delicious apples will need to be discarded because they are undersized.
 - a. What is the probability that an apple selected from a box will need to be discarded?
 - b. If ten apples are randomly selected from a box, find the probability that:
 - i. All the apples will have to be discarded
 - ii. Half of the apples will have to be discarded
 - iii. Less than two apples will be discarded

22. One bag contains three red and five white balls, and another bag contains four red and four white balls. One bag is chosen at random, a ball is selected from that bag, its colour is noted and then it is replaced.
- Find the probability that the ball chosen is red.
 - If the operation is carried out eight times, find the probability that:
 - Exactly three red balls are drawn
 - At least three red balls are drawn
23. a. If six die are rolled one hundred times, how many times would you expect the number of even numbers showing to exceed the number of odd numbers showing?
- b. If eight coins are tossed sixty times, how many times would you expect the number of heads to exceed the number of tails?

Answer:

- a. $\frac{1}{32}$; b. $\frac{5}{16}$; c. $\frac{5}{32}$; d. $\frac{31}{32}$
- ${}^6C_2 \left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)^2$
- a. ${}^7C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^4$; b. ${}^7C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^2$; c. $\left(\frac{1}{3}\right)^7$; d. $1 - \left(\frac{1}{3}\right)^7$
- a. ${}^4C_3 \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right)$; b. ${}^4C_2 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^2$
- $(0.65)^{12}$
- a. ${}^{12}C_3 \left(\frac{5}{6}\right)^9 \left(\frac{1}{6}\right)^3$; b. ${}^{12}C_8 \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right)^8$; c. ${}^{12}C_{10} \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^{10} + {}^{12}C_{11} \left(\frac{5}{6}\right) \left(\frac{1}{6}\right)^{11} + \left(\frac{1}{6}\right)^{12}$
- a. 0.2009; b. 0.7368; c. 0.2632
- $\left(\frac{5}{6}\right)^{15} + {}^{15}C_1 \left(\frac{5}{6}\right)^{14} \left(\frac{1}{6}\right) + {}^{15}C_2 \left(\frac{5}{6}\right)^{13} \left(\frac{1}{6}\right)^2$
- a. $\left(\frac{9}{10}\right)^{20} + {}^{20}C_1 \left(\frac{9}{10}\right)^{19} \left(\frac{1}{10}\right) + {}^{20}C_2 \left(\frac{9}{10}\right)^{18} \left(\frac{1}{10}\right)^2$; b. $1 - \left(\frac{9}{10}\right)^{20}$
- a. $0.91^{10} + {}^{10}C_1 \times 0.91^9 \times 0.09 + {}^{10}C_2 \times 0.91^8 \times 0.09^2$; b. $1 - 0.91^{10} - {}^{10}C_1 \times 0.91^9 \times 0.09$
- 0.593
- 0.000786
- ${}^{31}C_3 \times 0.95^{28} \times 0.05^3$
- a. i. 0.10764, ii. 0.11372; b. 0.78549
- a. 17; b. 7
- a. i. 0.487, ii. 0.031; b. We have assumed that boys and girls are equally likely. We have also assumed that in any one family, the events 'having a boy' and 'having a girl' are independent.
- a. The argument is invalid. Normally, mathematics books are grouped together, so once the shelf is chosen, one would expect all or none of the books to be mathematics books, thus the five stages are not independent events. The result would be true if the books were each chosen at random from the library.
- b. The argument is invalid. People in a particular neighbourhood tend to vote more similarly than the population at large, so the four events are not independent. The result would be true if one chose four streets at random, and then chose a voter randomly from each street.
- a. 0.03456; b. 0.68256
- a. 0.409600; b. 0.001126; c. 0.000869
- a. 0.0060; b. 0.0303
- a. $\frac{3}{250}$; b. i. $\left(\frac{3}{250}\right)^{10}$, ii. ${}^{10}C_5 \left(\frac{3}{250}\right)^5 \left(\frac{247}{250}\right)^5$, iii. $\left(\frac{247}{250}\right)^{10} + 10 \left(\frac{247}{250}\right)^9 \left(\frac{3}{250}\right)$
- a. $\frac{7}{16}$; b. i. ${}^8C_3 \left(\frac{9}{16}\right)^5 \left(\frac{7}{16}\right)^3$, ii. $1 - \left(\frac{9}{16}\right)^8 - 8 \left(\frac{9}{16}\right)^7 \frac{7}{16} - {}^8C_2 \left(\frac{9}{16}\right)^6 \left(\frac{7}{16}\right)^2$
- a. 34; b. 22